

WASSERKUPPE, Germany

WMO: 105440

Lat: 50.4972N

Lon: 9.9428E

Elev: 3037

StdP: 13.15

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.5	10.4	-1.3	5.7	12.5	2.9	7.1	14.7	41.4	31.8	37.2	27.5	15.4	130	0.809

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	75.3	62.0	72.1	60.5	69.3	58.9	63.7	72.4	61.8	70.1	60.0	67.4	11.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	87.3	68.0	58.4	81.7	65.2	56.9	77.2	63.1	30.6	72.0	29.2	70.3	27.8	67.2	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.1	30.1	26.9	DB	4.6	82.7	8.4	3.0	-1.5	84.9	-6.4	86.7	-11.1	88.4	-17.2	90.6	(4)
(5)			WB	3.9	67.9	8.1	1.6	-1.9	69.0	-6.6	70.0	-11.2	70.8	-17.0	72.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	27.3	26.4	32.5	38.6	48.2	53.0	57.6	57.6	50.9	43.8	34.5	30.2	(6)
(7)		DBStd	13.36	8.28	9.07	7.29	7.48	7.70	7.52	6.97	6.81	6.55	6.98	7.54	6.61	(7)
(8)		HDD50	3829	704	662	543	349	131	53	12	12	67	218	466	612	(8)
(9)		HDD65	8500	1169	1082	1006	793	521	366	245	244	423	659	915	1077	(9)
(10)		CDD50	842	0	0	1	6	75	143	249	248	94	25	1	0	(10)
(11)		CDD65	36	0	0	0	0	0	5	16	15	0	0	0	0	(11)
(12)		CDH74	156	0	0	0	0	0	14	60	82	0	0	0	0	(12)
(13)		CDH80	19	0	0	0	0	0	1	4	14	0	0	0	0	(13)
(14)	Wind	WSAvg	13.9	16.3	15.7	15.0	13.4	13.3	11.6	11.6	11.3	12.8	15.5	15.0	15.2	(14)
(15)	Precipitation	PrecAvg	34.6	3.2	2.7	2.5	2.0	2.9	2.8	3.5	2.9	2.7	2.8	3.1	3.5	(15)
(16)		PrecMax	42.9	7.3	7.6	5.1	3.8	6.4	6.7	6.2	7.7	6.3	9.1	4.9	9.7	(16)
(17)		PrecMin	24.1	0.1	0.8	0.6	0.1	0.4	1.0	1.3	0.7	0.7	0.5	0.1	0.8	(17)
(18)		PrecStd	5.2	1.6	1.5	1.3	0.9	1.6	1.3	1.2	1.4	1.4	1.7	1.3	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	50.4	58.0	61.6	70.7	75.9	79.7	81.9	72.0	66.4	55.6	47.9	(19)
MCWB			38.5	42.3	45.7	49.7	56.4	62.2	65.4	64.9	59.2	55.0	45.2	43.3	(20)	
2%		DB	41.7	45.0	50.1	57.0	66.9	72.0	75.3	75.6	68.1	60.6	50.2	44.2	(21)	
		MCWB	36.9	40.8	42.3	47.0	54.2	60.1	62.0	62.0	57.8	53.1	43.3	40.8	(22)	
5%		DB	38.8	41.1	45.6	53.6	63.8	68.2	71.9	71.9	64.6	56.4	47.3	41.9	(23)	
		MCWB	36.4	37.2	41.1	45.0	52.9	58.1	60.6	60.8	56.7	50.7	43.4	38.9	(24)	
10%		DB	36.9	37.4	41.8	49.8	60.2	64.6	68.9	68.2	60.7	53.1	44.7	39.0	(25)	
		MCWB	35.4	35.0	39.1	43.5	51.1	55.9	59.2	58.7	54.6	49.5	42.1	37.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	44.6	47.2	50.9	58.7	64.8	66.7	66.7	62.5	57.3	48.4	44.9	(27)
MCD B			43.2	47.0	53.0	59.0	65.9	73.8	76.6	78.8	69.0	63.7	50.6	45.8	(28)	
2%		WB	38.8	41.3	44.0	48.3	55.8	61.4	63.9	63.4	59.3	54.2	46.2	42.5	(29)	
		MCD B	39.5	44.2	48.9	55.0	63.8	69.2	72.2	73.0	65.2	58.1	48.1	43.7	(30)	
5%		WB	37.3	38.1	41.5	46.2	53.8	59.3	61.8	61.6	57.4	52.0	44.5	40.0	(31)	
		MCD B	37.9	40.3	44.5	51.7	62.1	66.1	69.9	70.2	61.7	55.3	46.5	41.1	(32)	
10%		WB	35.4	35.4	39.3	44.0	51.9	57.1	59.9	59.6	55.6	49.9	42.3	37.8	(33)	
		MCD B	36.5	36.8	41.7	49.4	59.1	63.4	67.4	66.6	59.8	52.4	44.2	38.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	8.2	8.6	11.5	13.0	12.1	13.2	12.9	11.1	9.5	7.5	7.2	(35)
MCDBR			11.2	13.3	14.0	16.6	18.0	16.7	17.1	18.0	15.8	14.6	11.6	10.5	(36)	
MCWBR			8.9	9.0	9.0	9.7	10.3	9.9	9.9	10.3	9.1	9.0	8.6	7.7	(37)	
5% WB		MCDBR	9.7	12.4	12.5	14.9	16.4	15.7	16.5	17.0	13.9	11.7	10.8	8.4	(38)	
		MCWBR	9.0	9.1	8.8	9.6	10.2	9.9	10.4	10.3	9.2	8.3	9.0	7.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.295	0.325	0.356	0.361	0.361	0.370	0.365	0.337	0.318	0.299	0.274	(40)	
taud		2.461	2.427	2.404	2.351	2.353	2.363	2.330	2.370	2.437	2.494	2.497	2.460	(41)		
Ebn at Noon		248	270	279	281	284	284	279	275	272	258	235	230	(42)		
Edh at Noon		18	24	30	36	37	37	38	35	29	23	18	16	(43)		
(44)	All-Sky Solar Radiation	RadAvg	243	470	818	1300	1536	1694	1605	1383	993	581	268	181	(44)	
(45)		RadStd	40	71	113	166	174	153	193	134	139	67	37	21	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+99	N/A

Nomenclature: See separate page